Suspected vitreous seeding of uveal melanoma: relevance of diagnostic vitrectomy

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ABSTRACT

Aim To review all cases of suspected vitreous seeding of treated or untreated uveal melanoma at our clinic and to compare clinical, cytological and histological findings with patients' survival.

Methods Retrospective non-randomised study of 23 patients with consecutive uveal melanoma who underwent diagnostic vitrectomy in our clinic between January 2000 and November 2013. Reason for vitrectomy was suspected dissemination of tumour cells inside the eye. Treated as well as treatment-naïve primary uveal melanomas were included in this study. Follow-up data of all patients were collected.

Results The study included 23 patients with a mean age of 66 years. Four patients presented pigmented vitreous debris at initial presentation prior to treatment of the uveal melanoma. All but one of these four patients has been enucleated as a consequence of cytology-proven vitreous spreading of vital melanoma cells. The remaining 19 patients presented pigmented vitreous debris at a mean of 60 months following local tumour treatment. Thirteen of these patients had been treated with a ruthenium plaque (mean scleral dose 1295 Gy, mean apex dose 152 Gy), three with binuclid plaque (mean scleral dose 1005 Gy, mean apex dose 70 Gy) and three with proton beam radiation. Of the 19 patients, 10 showed only melanophages in the vitreous specimen, while the remaining 9 patients had vital tumour cells in vitreous cytology. Four out of these nine patients have been enucleated in the course of follow-up. During follow-up of our cohort of 23 patients, 4 patients died, but only 1 of them due to metastatic disease.

Conclusion The outcome of this small cohort study shows that obtaining a vitreous specimen helps to distinguish melanophages from vital tumour cells. We could not observe an increased risk of metastasis in patients who showed melanoma cell dissemination inside the eye, compared with those patients only showing melanophages. We therefore suggest to carefully re-evaluate the necessity of enucleation in every patient.

INTRODUCTION

Uveal melanoma is the most common primary intraocular tumour with a reported incidence of seven out of one million people.[1] In very few cases, patients with uveal melanoma develop vitreous seeding, and most of these tumours have a characteristic mushroom shape.[2] First mentioned by Knapp in 1864 and further analysed by Ronne in the early 20th century, Jensen[3] coined the term Knapp-Ronne-type uveal melanoma in 1976. Knapp-Ronne-type uveal melanomas grow fast and are usually located close to the optic disc where the retina is tighter. Due to the rapid tumour growth, the overlying retina becomes atrophic and Bruch's membrane strangulates the tumour. This causes vitreous bleeding in some cases, as reported by Gass.[4]

In 1988, Dunn *et al* analysed one case of vitreous seeding without vitreous bleeding. They found thinning of the retina overlying the tumour, as well as asteroid bodies with surrounding tumour cells in a collarette pattern close to the tumour surface.[2] Diagnostic vitrectomy has been employed since the 1980s and has become increasingly important.[5 6] In 1987, Traboulsi *et al*[7] already described one case of a patient who presented chronic vitreous haemorrhage, and although not expected, the vitreous cytology probe of this patient showed melanoma cells. Papanicolaou stain showed cells that were variable in size and shape with eccentric nuclei and prominent nucleoli. Melanin granules were small and uniform and evenly dispersed in the cytoplasm.[7] Pigment dispersion throughout the vitreous might be caused by tumour necrosis[8] and does not necessarily indicate tumour spreading. After brachytherapy, pigmented cells might shed into the vitreous, and aggregates of pigmented cells might indicate vital tumour cells.[9]

In our study, we analysed all cases of suspected vitreous tumour seeding to find out if there is a negative association with patients' tumour-related survival. Vitrectomies have been performed due to suspected uncontrolled tumour disease, some of the patients complained visual disturbances and some patients addressed the fear of uncontrolled tumour disease.

PATIENTS AND METHODS

This retrospective non-randomised study includes 23 consecutive patients who underwent diagnostic vitrectomy at our clinic between January 2000 and November 2013 due to suspected vitreous seeding of uveal melanoma cells. Inclusion criterion was evidence of pigment inside the vitreous without prior diagnostic vitrectomy/tumour biopsy. Patients were either treated with a ruthenium or binuclid plaque, proton beam radiation or they were treatment-naïve. We excluded patients who previously had a diagnostic vitrectomy, tumour biopsy or transretinal endoresection.

Data collection

Baseline data collection included age, sex, tumour location, largest basal diameter and tumour height. Follow-up data included latency between local tumour treatment and diagnostic vitrectomy, cytological data of vitreous specimen, histological findings in case of enucleation, scleral and apex dosage in case of ruthenium or binuclid plaque or overall radiation dose after proton beam radiation. Follow-up data concerning tumour-related survival were collected.

RESULTS

The study included 23 patients (13 female and 10 male) with a mean age of 66 years (range 34–84 years) at clinical diagnosis of

Table 1 Patient cohort characteristics

Baseline characteristics	Value			
Total number of patients n (%)	23 (100)			
Sex n (%)				
Male	10 (43)			
Female	13 (57)			
	Value	**Range**		
Age (mean) in years (SD)	65.56 (14.33)	34–84		
Mean baseline visual acuity (logMAR) (SD)	0.77 (0.7)	0.1–2.0		
Mean tumour height in mm (SD)	6.5 (2.99)	2.3–12.3		
Mean LBD in mm (SD)	11.4 (3.12)	6.28–17.3		
	Value			
Tumour location	n (%)			
Temporal	12 (52.2)			
Nasal	2 (8.7)			
Inferior	2 (8.7)			
Superior	0 (0)			
Juxtapapillary	4 (17.4)			
Posterior pole	3 (13.0)			
Ciliary body involvement				
Yes	4 (17.4)			
No	19 (82.6)			
Tumour height (mm)	n (%)			
<3	4 (17.4)			
3–6	7 (30.4)			
6.1–9	5 (21.7)			
9.1–15	6 (26.1)			
>15	0 (0)			
LBD (mm)	n (%)			
<3	0 (0)			
3.1–12.0	12 (52.2)			
12.1–15	3 (13.0)			
15.1–18	3 (13.0)			
>18	0 (0)			
Unknown	5 (21.7)			
Seventh TNM classification TNM category	**Subclassification a**	**Subclassification b**	**Subclassifications c–d**	
T1	3 (13%)	–	–	
T2	5 (21.7%)	3 (13.0%)	–	
T3	6 (26.1%)	1 (4.3%)	–	
T4	–	–	–	
Unknown	5 (21.7%)			
	Vitreous specimen showing only melanophages		**Vitreous specimen showing tumour cells**	
	Value	**Range**	**Value**	**Range**
Treated cohort				
Number of patients n (%)	10 (43.5)		9 (39.1)	
Age (mean) in years (SD)	62.88 (13.02)	46.3–81.4	66.2 (14.73)	33.4–77.6
Mean LBD in mm (SD)	13.62 (2.93)	9.2–17.3	10.41 (2.28)	7.8–14.3
Mean tumour height in mm (SD)	6.6 (3.1)	3–11.1	5.5 (2.8)	2.3–10.2

Continued

Table 1 Continued

Baseline characteristics	Value	Range	Value	Range
Mean time interval between treatment of primary tumour and diagnostic vitrectomy in months	77.58	2.3–251.2	54.4	7.4–131
	Vitreous specimen showing only melanophages		**Vitreous specimen showing tumour cells**	
	Value	**Range**	**Value**	**Range**
Brachytherapy—Ruthenium (n=13)				
Number of patients n	7		6	
Mean scleral dose in Gy	1173	700–1250	1312	700–1874
Mean apex dose in Gy	104	70–158	167	100–328
Brachytherapy–Binuclid plaque (n=3)				
Number of patients n	2		1	
Mean scleral dose in Gy (range)	1005 (928–1063)			
Mean apex dose in Gy (range)	70 (–)			
	Value		**Value**	
Proton beam radiation (n=3)				
Number of patients n (%)	1		2	
Mean radiation dose in Gy	60		60	
	Value	**Range**	**Value**	**Range**
Treatment-naïve cohort				
Number of patients n (%)	0	–	4 (17.4)	
Age (mean) in years (SD)	–	–	70 (19.62)	41–84.3
Mean LBD in mm (SD)	–	–	9.05 (2.55)	6.28–11.8
Mean tumour height in mm (SD)	–	–	7.73 (3.3)	4.3–12.3

LBD, largest basal diameter; TNM, tumour, node, metastases.

uveal melanoma (see table 1). Mean follow-up time after diagnostic vitrectomy was 309 days (range 2–1236 days).

In 4 out of 23 cases, pigment dispersion was present without prior tumour treatment at initial presentation at our clinic (figure 1). Diagnostic 25-gauge vitrectomy was performed and followed by enucleation in three of these four cases due to cytology-proven diffuse spreading of the tumour. In the fourth case, we decided to perform a transretinal endoresection after adjuvant Gamma Knife radiation instead of enucleation, and good tumour control could be achieved (follow-up time 37 months). None of these four patients died of a tumour-related cause. In the treatment-naïve patient cohort of four patients, the location of the tumour was juxtapapillary in one case, was in the mid-periphery in one case and was in peripheral of the equator in two other cases. In one of the four cases a ciliary body involvement was observed. Mean patient age was

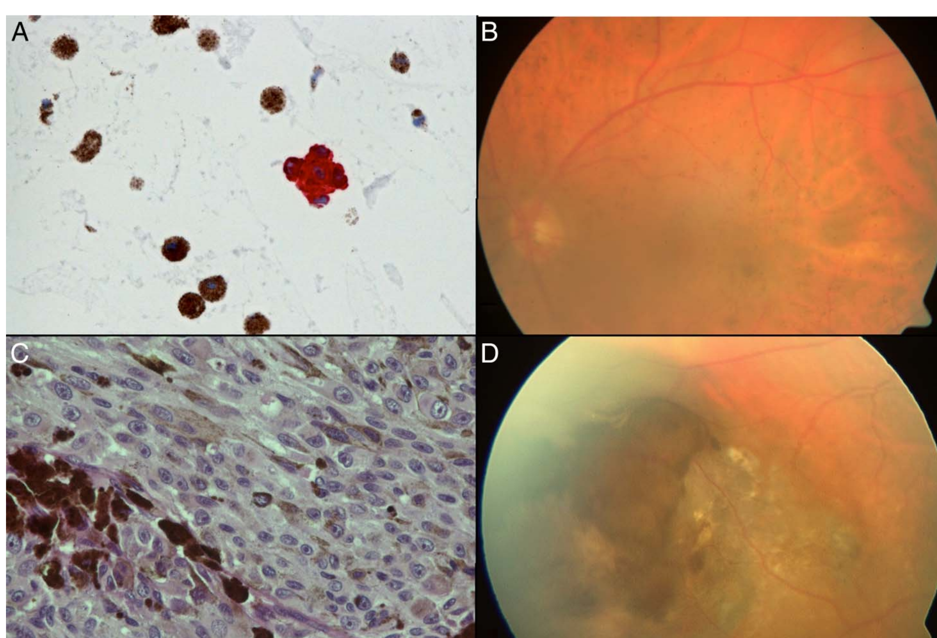

Figure 1 Patient without prior treatment. (A) Cytology Melan-A staining with red tumour cells. (B) Disseminated pigmentation of the fundus. (C) Histology of enucleated eye (H&E staining). (D) This uveal melanoma shows infiltration of the retina.

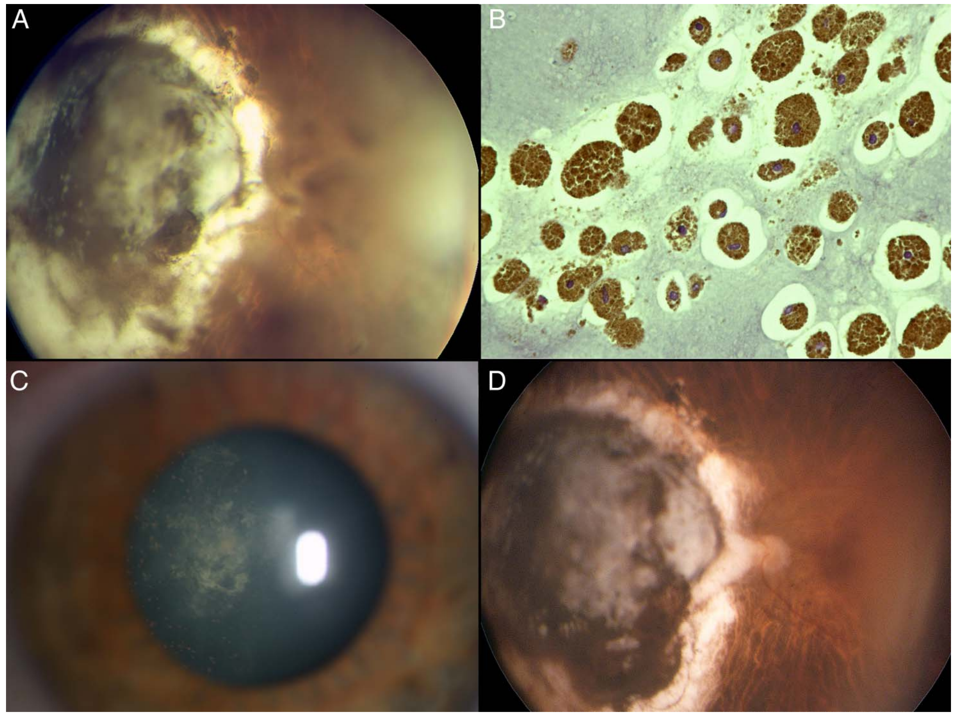

Figure 2 Uveal melanoma within the scar 7 years after brachytherapy. (A) Disseminated pigmentation in the vitreous body. (B) Cytology shows melanophages, without melanoma cells (H&E staining). (C) Slit lamp picture prior to vitrectomy showing pigmented conglomerates behind the lens. (D) Fundus picture 3 years after vitrectomy.

70 years (range 41–84 years) at the time of detection of suspected vitreous seeding. Mean follow-up after biopsy was 355 days. A transretinal endoresection was performed in one case and good visual acuity (0.63 Snellen equivalent or logMAR 0.2) could be preserved in this case. Mean tumour height was 7.7 mm (range 4.3–12.3 mm) with a largest basal diameter of 9 mm (range 6.3–11.8 mm). In all four cases, histopathology revealed vital tumour cells.

The remaining 19 patients showed pigmented debris in the vitreous cavity after primary tumour treatment (figures 2–4). Therefore, a diagnostic vitrectomy was performed in mean 60 months after tumour treatment. Postoperative follow-up ranged from 2 days to 41 months (mean 22 months). We performed a vitrectomy in every case and in 10 cases cytology revealed melanophages without any vital tumour cells (exemplary patient, figure 3). The other nine patients showed vital tumour cells in the cytology specimen, and enucleation was performed after proof of vital tumour disease (figures 3 and 4). Average tumour height was 6.6 mm (SD 3.1) (range 3–11.1 mm) in the cohort of the 10 patients that showed melanophages in

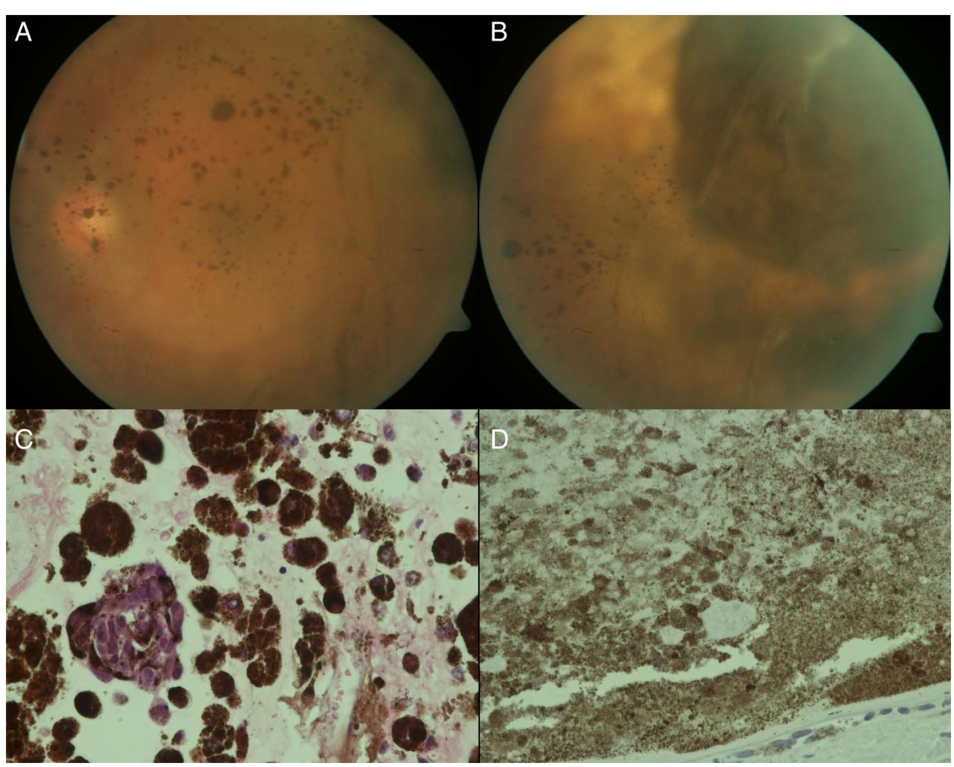

Figure 3 (A) Pigmented conglomerates throughout the vitreous body. (B) Thin overlying retina origin of pigmented conglomerates. (C) Cytology (H&E staining) of melanoma cell conglomerates and melanophages. (D) Melan-A-negative histology after enucleation showing only necrotic cells.

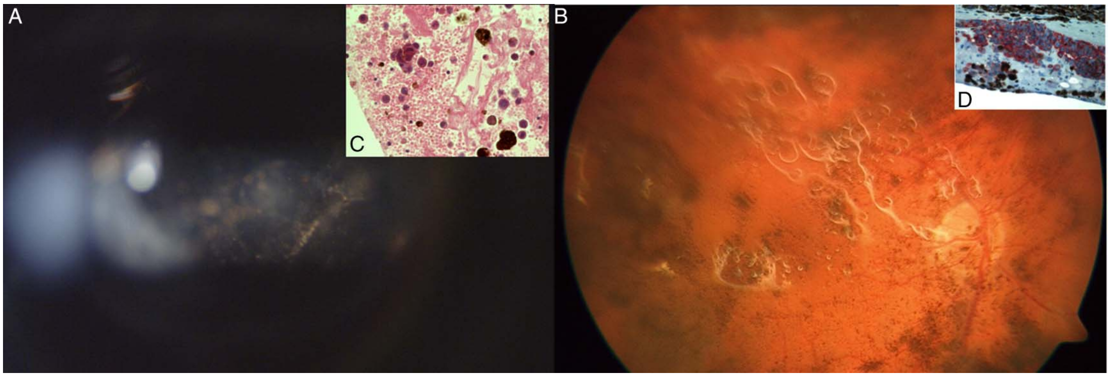

Figure 4 (A) Slit lamp picture 9 years after brachytherapy. (B) Fundus picture after pars plana vitrectomy and silicon oil tamponade showing disseminated pigmentation throughout the fundus. (C) Cytology (H&E staining). (D) Histology of the enucleated eye, positive (red) for Melan-A.

vitreous specimen and 5.5 mm (SD 2.8) (range 2.3–10.2 mm) in the cohort of the 9 patients that showed vital tumour cells in vitreous specimen (see table 1). The location of the tumours varied. The tumour had ciliary body involvement in three cases, juxtapapillary location in three further cases and in the remaining three cases it was at the posterior pole. The other tumours were located in the mid-periphery.

Of the 19 patients, 4 were treated twice with a ruthenium plaque, 9 patients were treated once with a ruthenium plaque, 3 patients were treated with a binuclid (ruthenium/iodine) plaque and 3 were treated with proton beam radiation. The latency between primary tumour treatment and necessity for diagnostic vitrectomy was on average 54 months (range 7–131 months) in the cohort of patients that showed vital tumour cells in vitreous biopsy. On average 78 months (range 2–251 months) passed in the cohort of patients whose vitrectomy showed merely melanophages.

The location of the tumour varied with no predominance for a certain location, and histopathology revealed highly pigmented and non-pigmented cells.

DISCUSSION

As far as we know, this retrospective consecutive case series shows the largest case series of patients with suspected vitreous seeding of uveal melanoma. The affected eye can only rarely be saved in cases of untreated uveal melanoma and vitreous pigment opacities. In our cohort, vitreous biopsy revealed vital tumour cells in all of these treatment-naïve cases. Enucleation often becomes necessary to provide complete tumour control in this cohort. We expected the juxtapapillary region to be the most common location for uveal melanoma with vitreous seeding, but no predominance for this location could be observed.

After brachytherapy, diagnostic vitrectomy evenly showed melanophages as well as vital melanoma cells. Possible reasons for diffuse vitreous pigment seeding might be the induction of posterior vitreous detachment induced via brachytherapy or shedding of cells into the vitreous after the overlying retina has become thinner. Necrosis of the overlying retina might be one reason for this observation.

In our treated cohort, no prevalence of juxtapapillary tumours could be seen. Clinical features do not give useful information to distinguish vital tumour cells from melanophages in the vitreous cavity. When diffuse pigmented vitreous opacities have been detected in the vitreous cavity, a diagnostic vitrectomy needs to be employed to differentiate melanophages from vital melanoma cells.

Only one patient out of our 23 patient cohort died due to a tumour-related cause, although more than half of our cohort (13/23 patients) had cytology-proven vital tumour cells in vitreous cavity. Moreover, we performed enucleations on two patients whose cytology showed vital melanoma cells in vitreous specimen, but the histology of the enucleated eye showed only necrotic tumour. Therefore, we think that the necessity for enucleation in patients with vitreous seeding should be carefully re-evaluated.

To conclude, in our small cohort we could not observe an increased risk of metastasis in patients who showed melanoma cell dissemination inside the eye, compared with those patients only showing melanophages. We think that the biology of the uveal melanoma (eg, chromosome 3 status or gene expression profiling) determines patients' tumour-related survival much more than possible vitreous spreading of the tumour inside the eye.

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Contributors CHDM and MG reviewed the literature, collected the data, participated in the analysis of data and prepared the manuscript. KAM and NB participated in the analysis of data and prepared the manuscript.

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REFERENCES

1 Damato B. Does ocular treatment of uveal melanoma influence survival? *Br J Cancer* 2010;103:285–90.
2 Dunn WJ, Lambert HM, Kincaid MC, *et al*. Choroidal malignant melanoma with early vitreous seeding. *Retina* 1988;8:188–92.
3 Jensen OA. The "Knapp-Ronne" type of malignant melanoma of the choroid. A haemangioma-like melanoma with a typical clinical picture. So-called "preretinal malignant choroidal melanoma". *Acta Ophthalmol (Copenh)* 1976;54:41–54.
4 Gass JD. Hemorrhage into the vitreous, a presenting manifestation of malignant melanoma of the choroid. *Arch Ophthalmol* 1963;69:778–9.
5 Engel HM, Green WR, Michels RG, *et al*. Diagnostic vitrectomy. *Retina* 1981;1:121–49.
6 Green WR. Diagnostic cytopathology of ocular fluid specimens. *Ophthalmology* 1984;91:726–49.
7 Traboulsi EI, Jalkh AE, Frangieh GT, *et al*. Vitreous histocytology of primary choroidal malignant melanoma. *Ann Ophthalmol* 1987;19:45–7.
8 El Baba F, Hagler WS, De la Cruz A, *et al*. Choroidal melanoma with pigment dispersion in vitreous and melanomalytic glaucoma. *Ophthalmology* 1988;95:370–7.
9 Robertson DM, Campbell RJ. Intravitreal invasion of malignant cells from choroidal melanoma after brachytherapy. *Arch Ophthalmol* 1997;115:793–5.